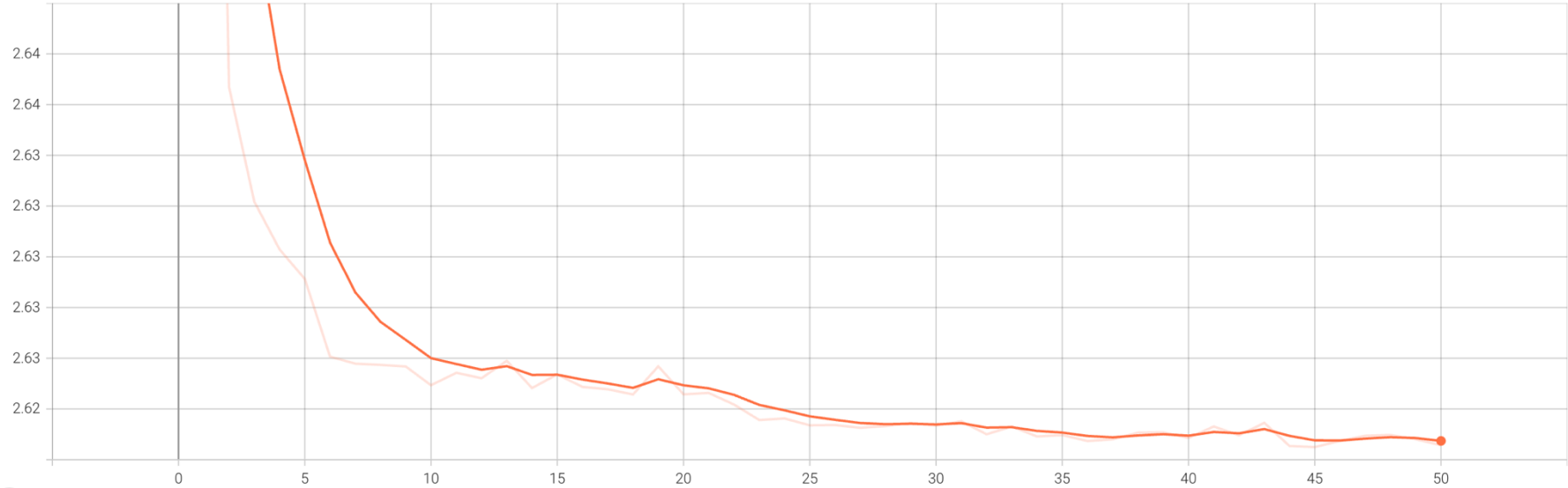


Graphsage neural network pre-training results

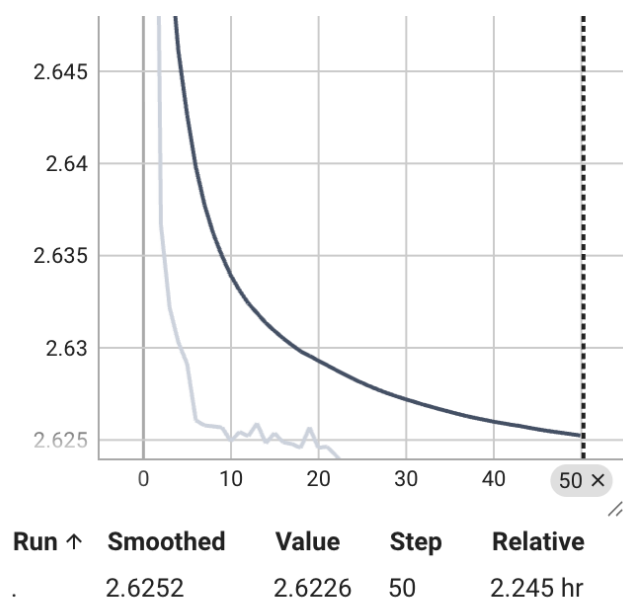
All_data dataset Graphsage pre-training

- 1. Dataset size
RNA nodes: 11854; Protein nodes: 237; edges: 226683
- 2. Parameter variables:
Positive pairs WALK_LEN = 1
Negative pairs N_WALK_LEN = 5
Batchsize = 128

Train/EpochLoss
tag: Train/EpochLoss



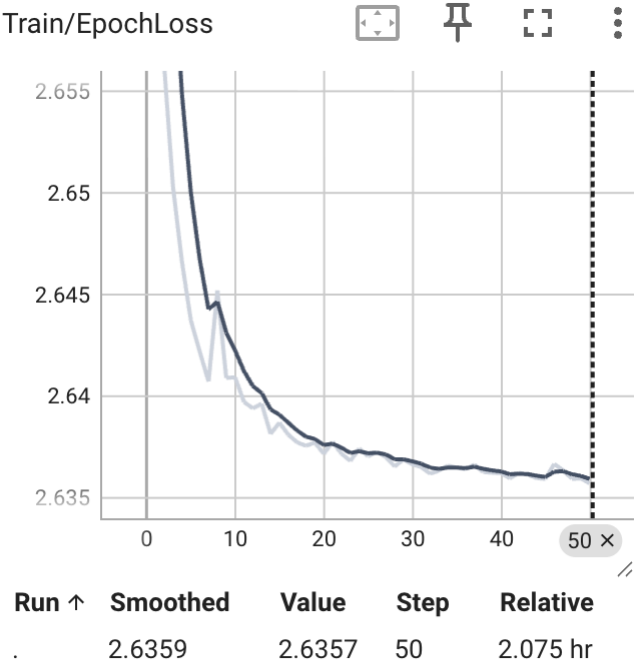
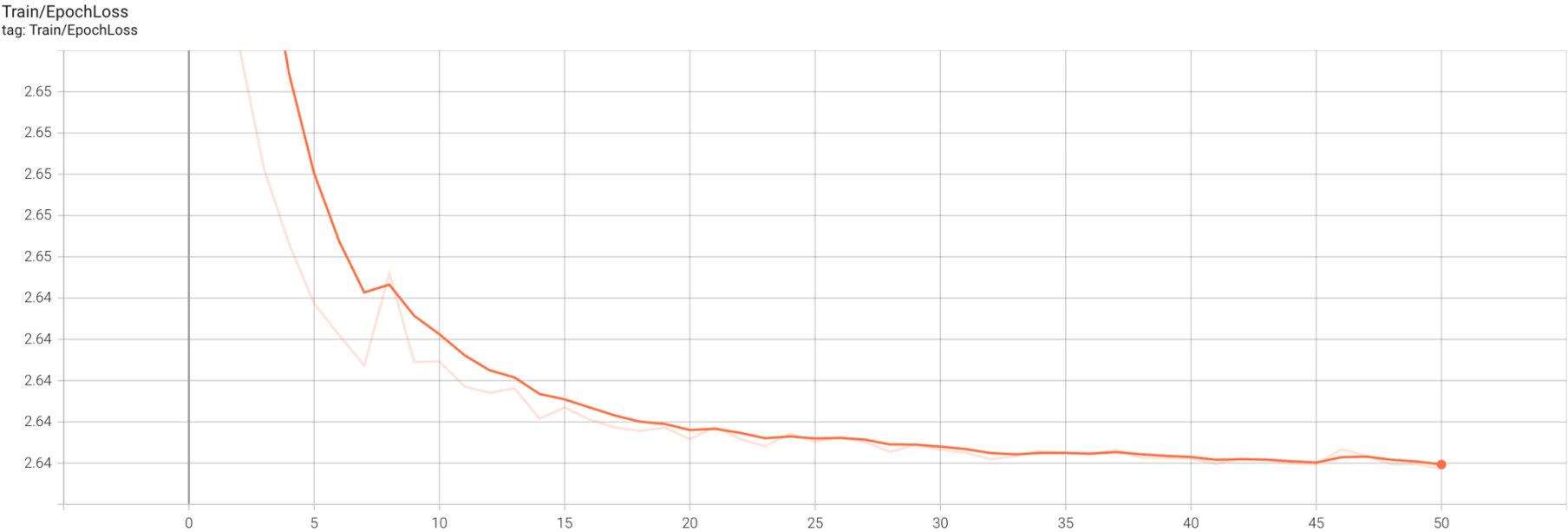
Train/EpochLoss



Run ↑	Smoothed	Value	Step	Relative
.	2.6252	2.6226	50	2.245 hr

All_data dataset Graphsage pre-training

- 1. Dataset size
RNA nodes: 11854; Protein nodes: 237; edges: 226683
- 2. Parameter variables:
Positive pairs WALK_LEN = 1
Negative pairs N_WALK_LEN = 4
Batchsize = 128



All_data dataset Graphsage pre-training

1. Dataset size

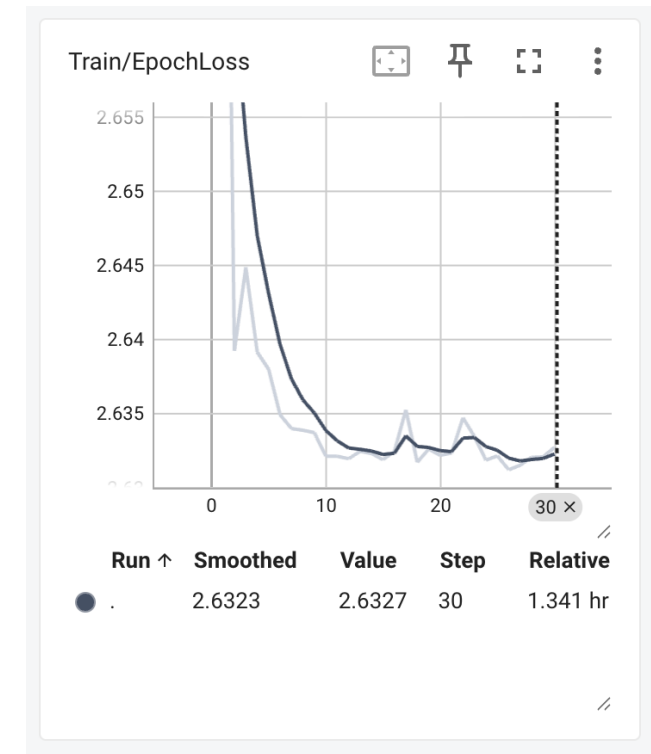
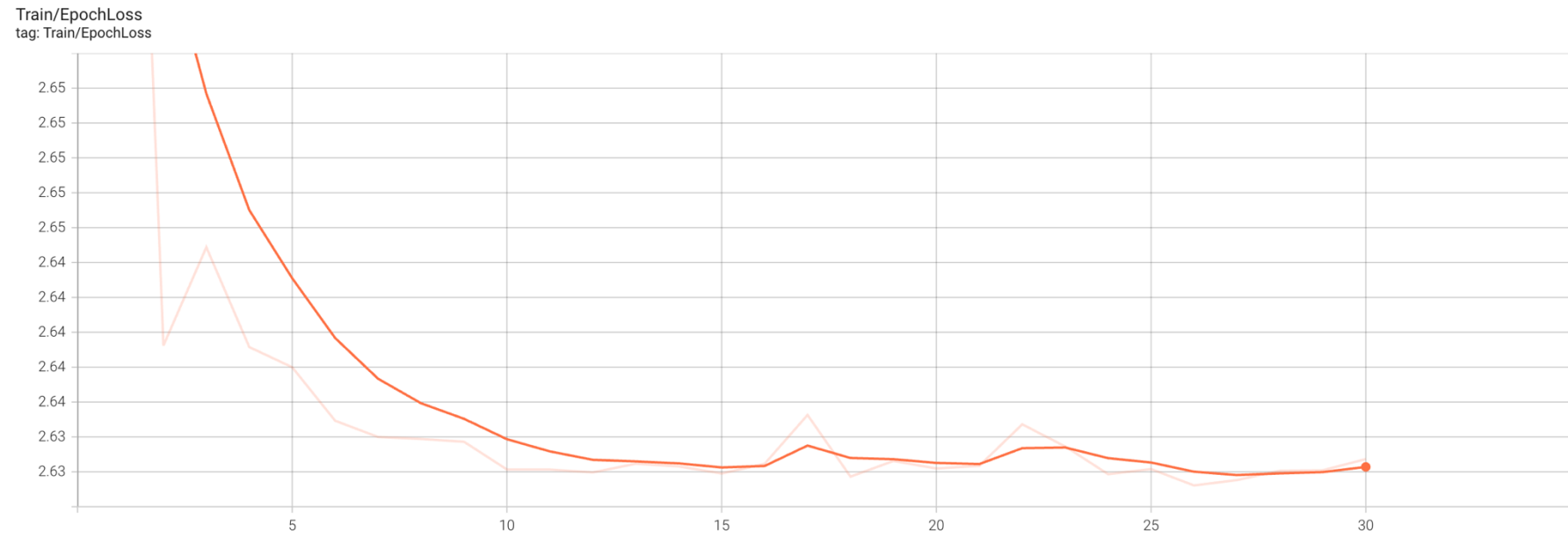
RNA nodes: 11854; Protein nodes: 237; edges: 226683

2. Parameter variables:

Positive pairs WALK_LEN = 2

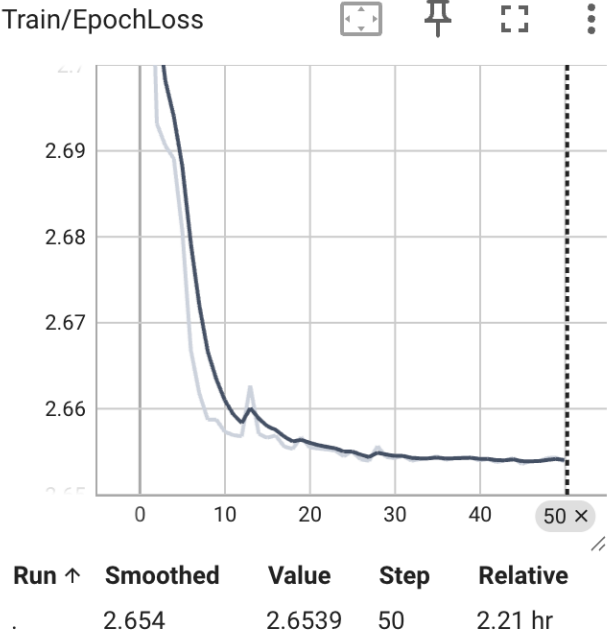
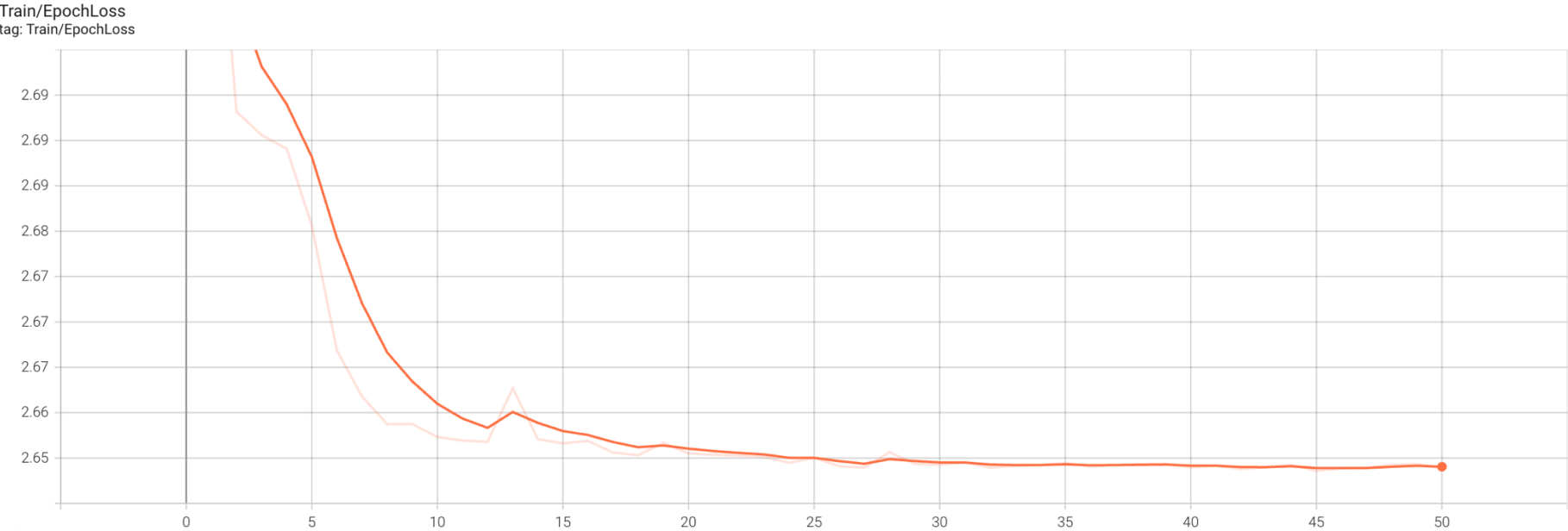
Negative pairs N_WALK_LEN = 5

Batchsize = 128



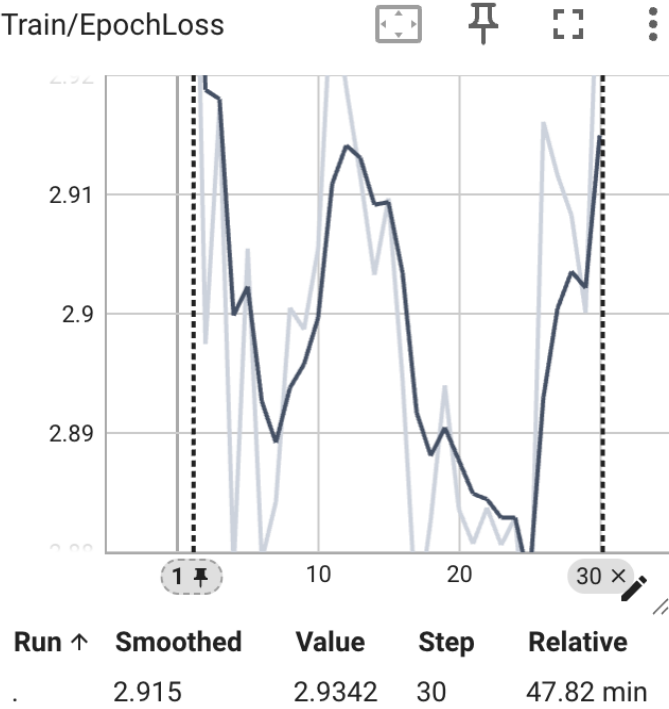
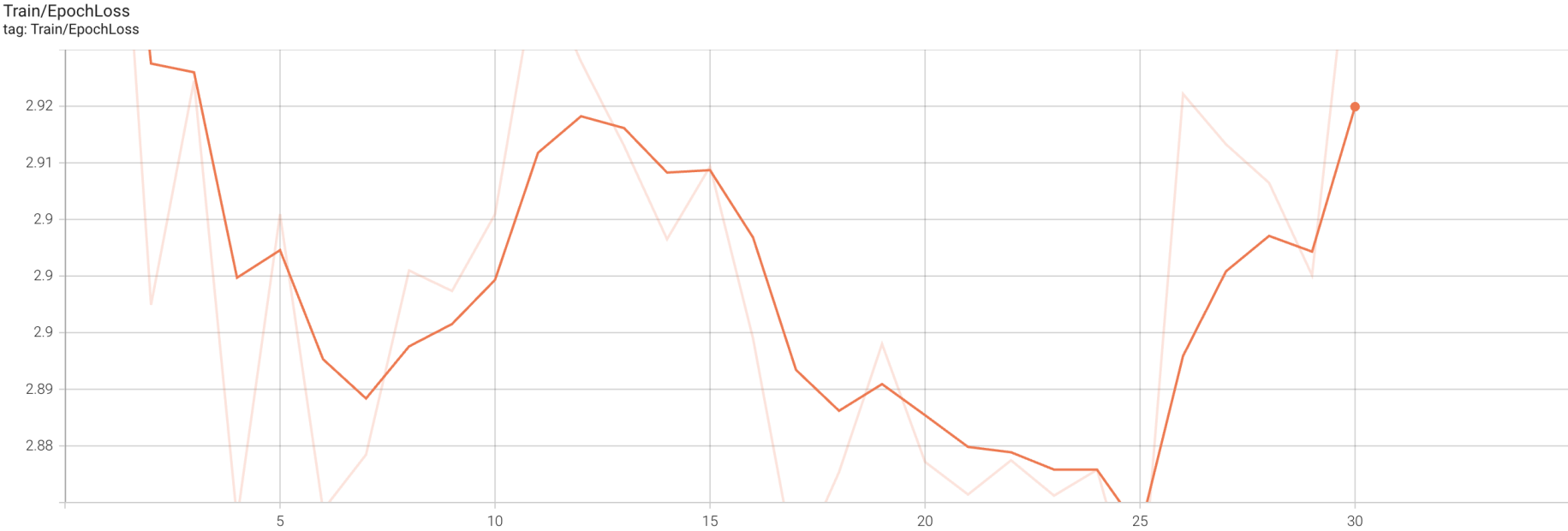
All_data dataset Graphsage pre-training

- 1. Dataset size
RNA nodes: 11854; Protein nodes: 237; edges: 226683
- 2. Parameter variables:
Positive pairs WALK_LEN = 2
Negative pairs N_WALK_LEN = 4
Batchsize = 128



All_data dataset Graphsage pre-training

- 1. Dataset size
RNA nodes: 11854; Protein nodes: 237; edges: 226683
- 2. Parameter variables:
Positive pairs WALK_LEN = 1
Negative pairs N_WALK_LEN = 3
Batchsize = 128



All_data dataset Graphsage pre-training

1. Dataset size

RNA nodes: 11854; Protein nodes: 237; edges: 226683

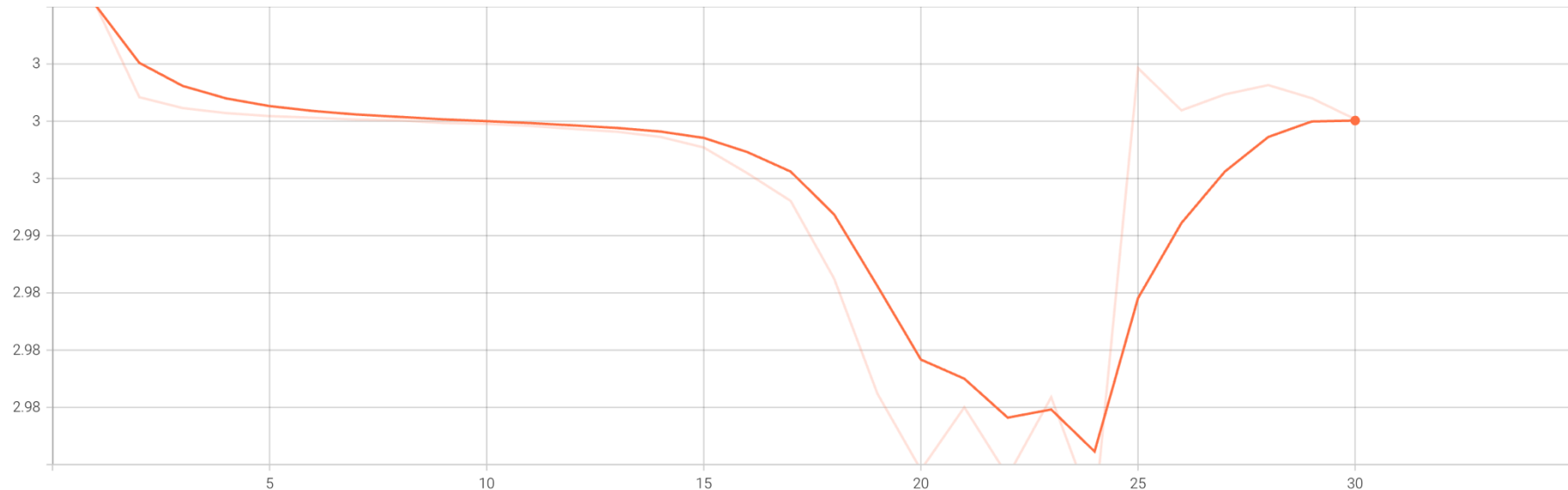
2. Parameter variables:

Positive pairs WALK_LEN = 2

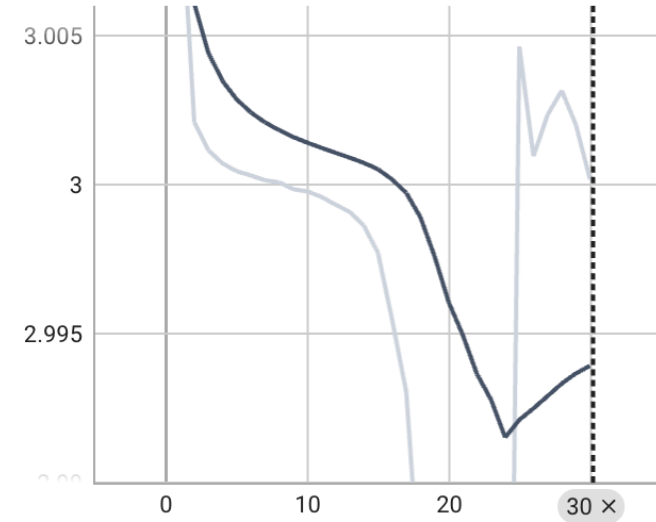
Negative pairs N_WALK_LEN = 3

Batchsize = 128

Train/EpochLoss
tag: Train/EpochLoss



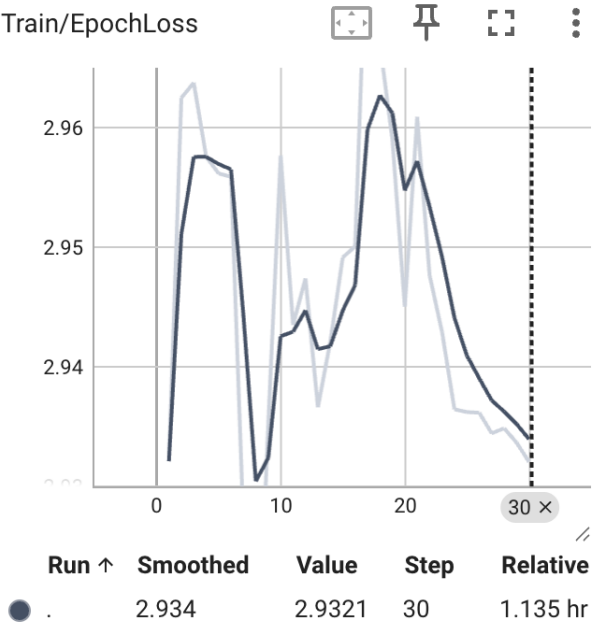
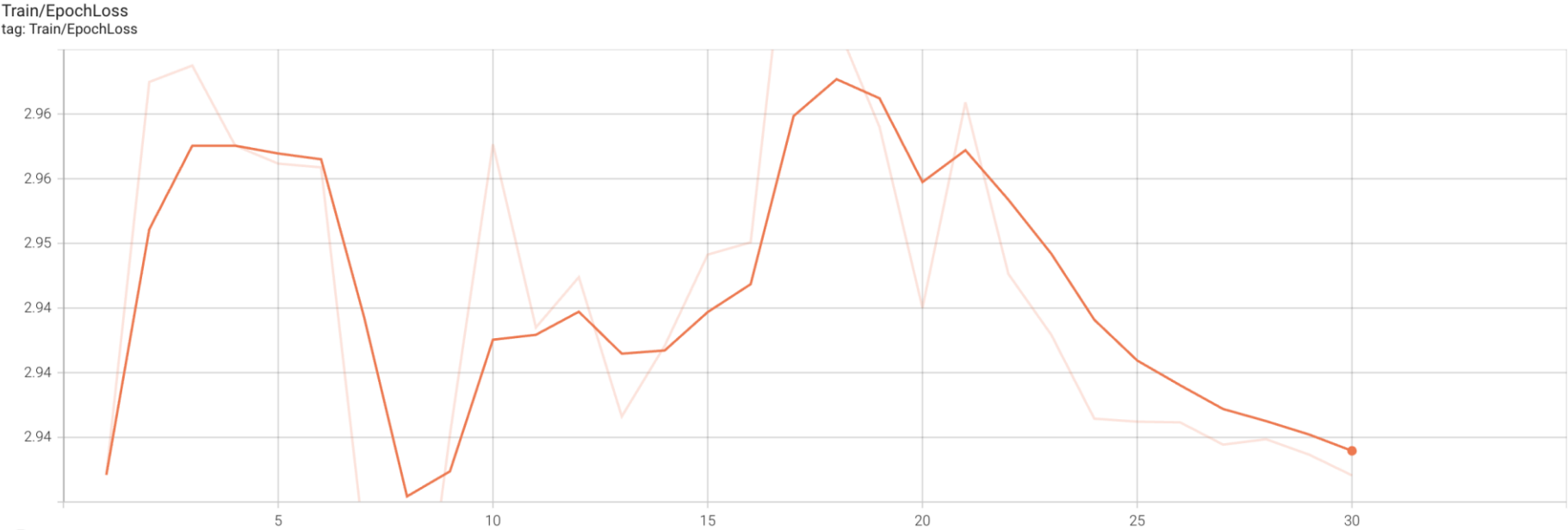
Train/EpochLoss



Run	Smoothed	Value	Step	Relative
1	2.9939	3.0002	30	50.94 min

All_data dataset Graphsage pre-training

- 1. Dataset size
RNA nodes: 11854; Protein nodes: 237; edges: 226683
- 2. Parameter variables:
Positive pairs WALK_LEN = 1
Negative pairs N_WALK_LEN = 3
Batchsize = 20



RPI7317 dataset Graphsage pre-training

1. Dataset size

RNA nodes: 1871; Protein nodes: 118; edges: 14634

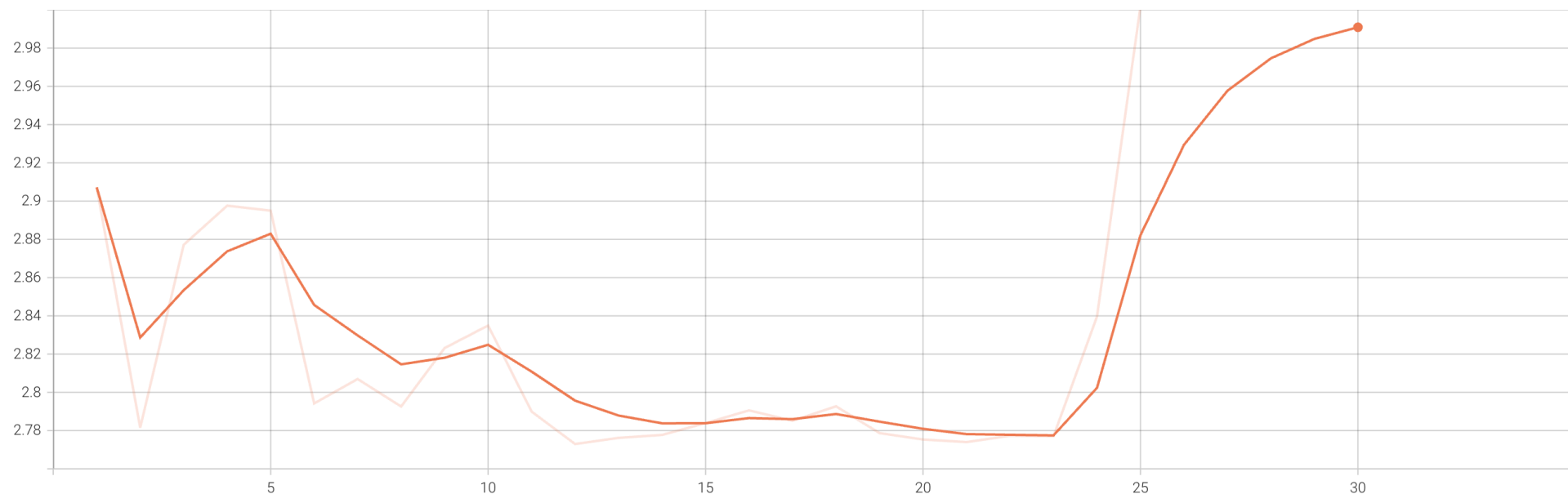
2. Parameter variables:

Positive pairs WALK_LEN = 1

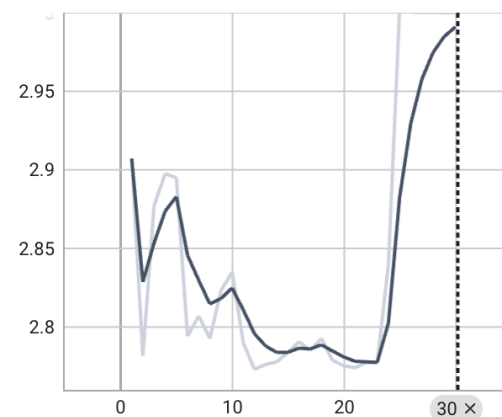
Negative pairs N_WALK_LEN = 3

Batchsize = 20

Train/EpochLoss
tag: Train/EpochLoss



Train/EpochLoss



Run ↑	Smoothed	Value	Step	Relative
.	2.9909	3	30	5.276 min

VecNN training results

VecNN training results

Train and test epoch loss of 5-fold cross-validation

Fold_0

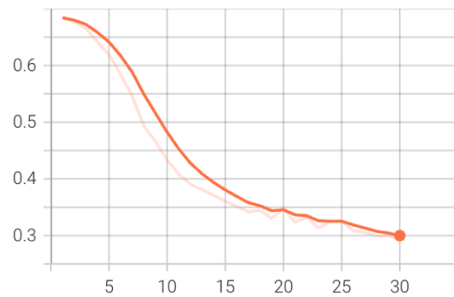
Fold_1

Fold_2

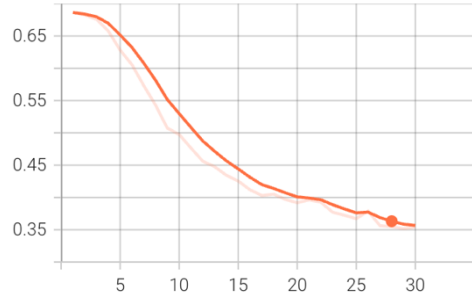
Fold_3

Fold_4

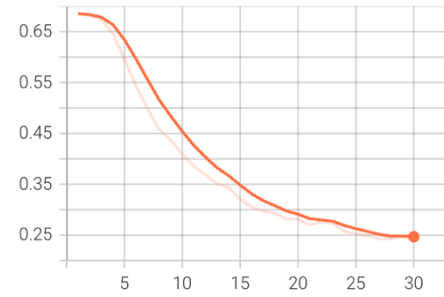
Train/EpochLoss
tag: Train/EpochLoss



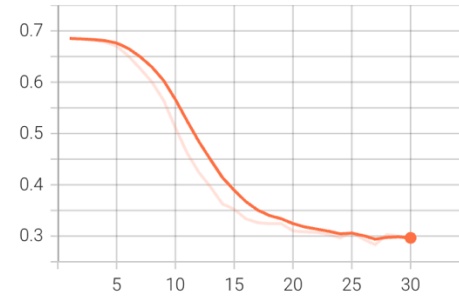
Train/EpochLoss
tag: Train/EpochLoss



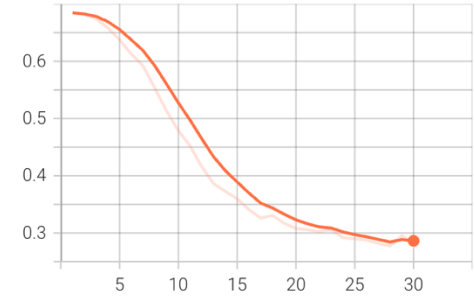
Train/EpochLoss
tag: Train/EpochLoss



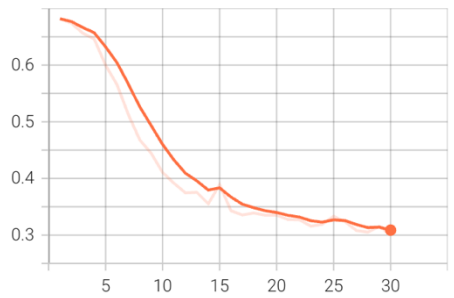
Train/EpochLoss
tag: Train/EpochLoss



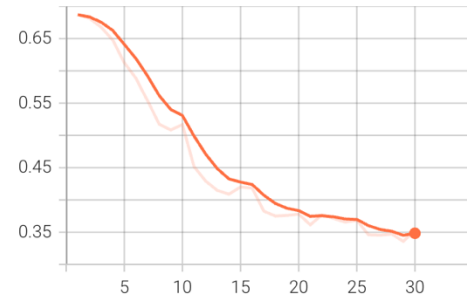
Train/EpochLoss
tag: Train/EpochLoss



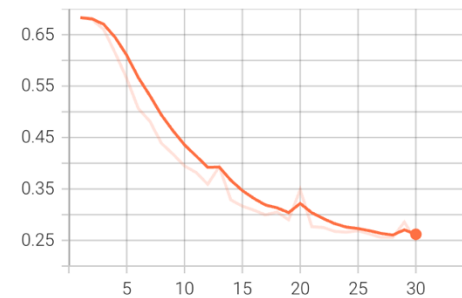
Val/Loss
tag: Val/Loss



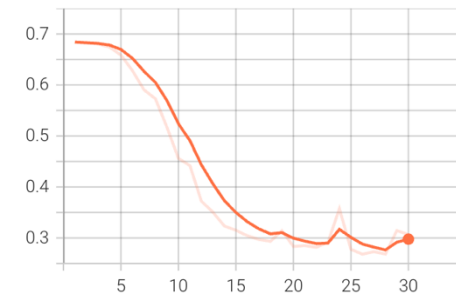
Val/Loss
tag: Val/Loss



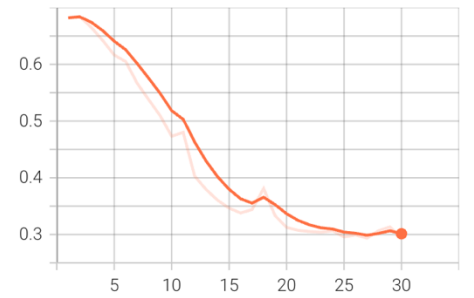
Val/Loss
tag: Val/Loss



Val/Loss
tag: Val/Loss

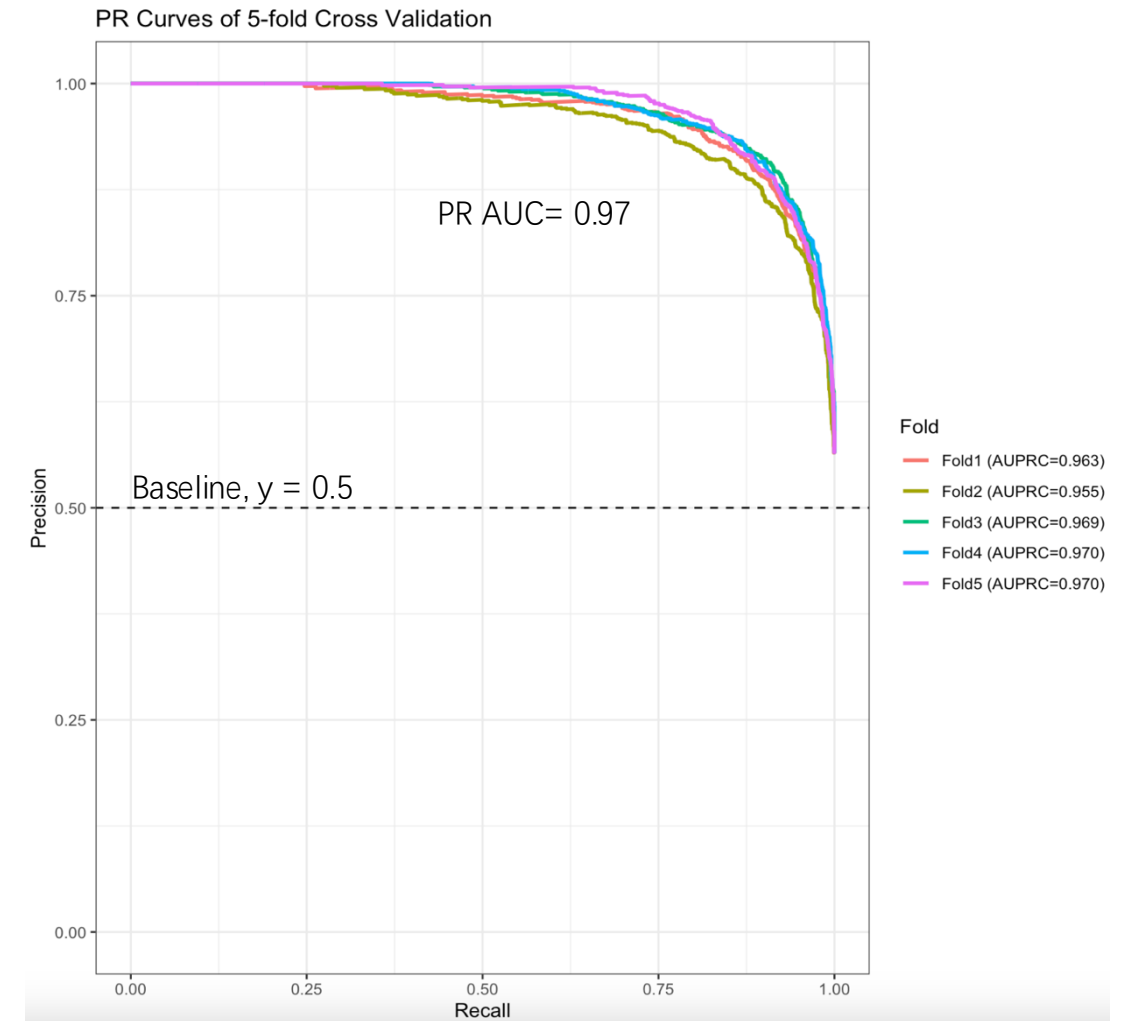
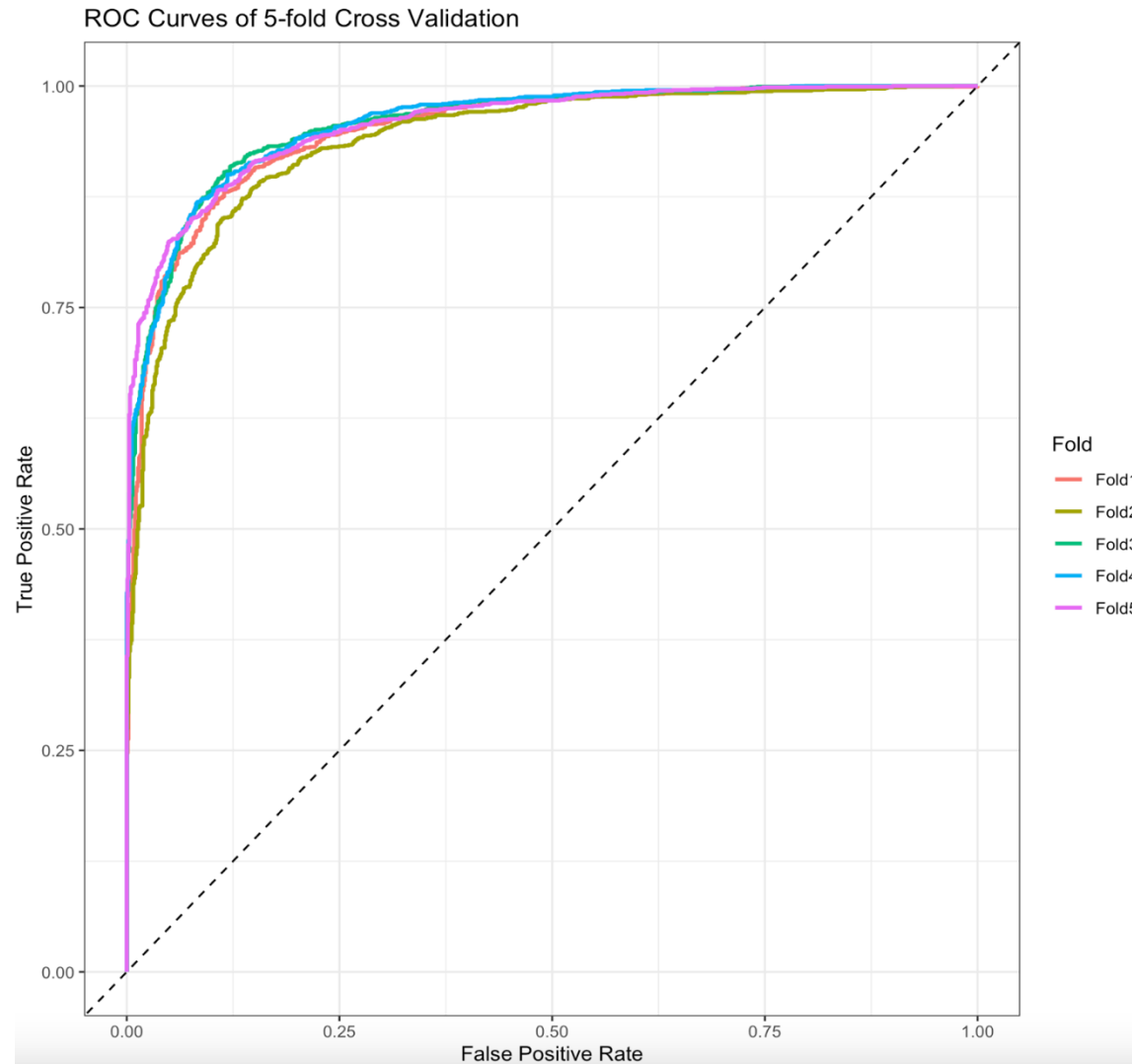


Val/Loss
tag: Val/Loss



VecNN training results

ROC and precision-recall curves of 5-fold cross-validation



VecNN training results

Performance metrics (AUC, AUPRC, ACC, Precision, Recall, Specificity, MCC) of the VecNN model

